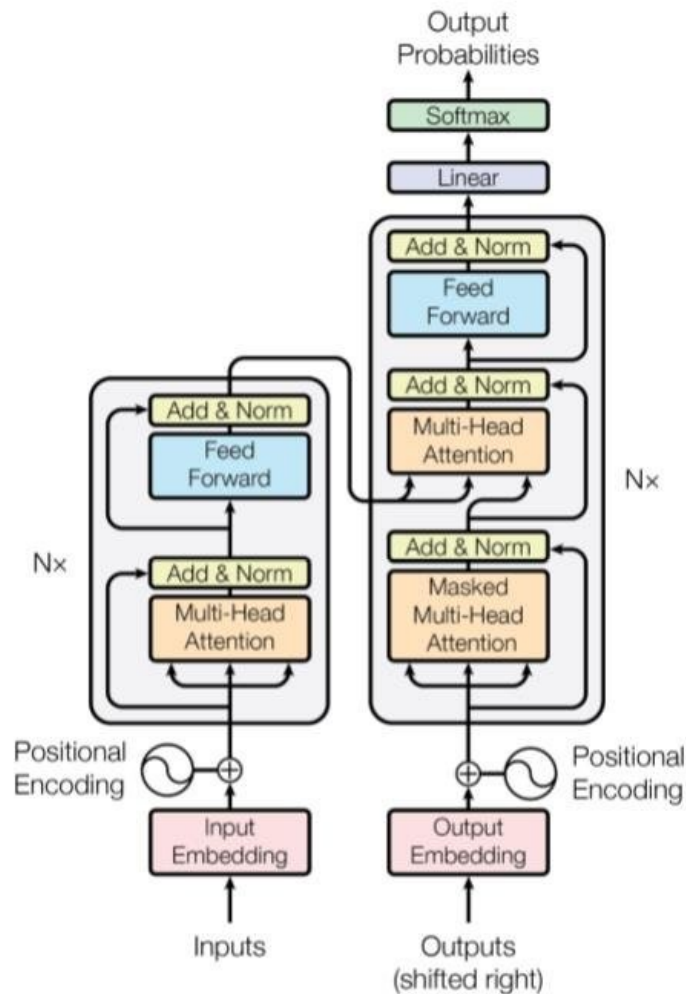


Трансформер

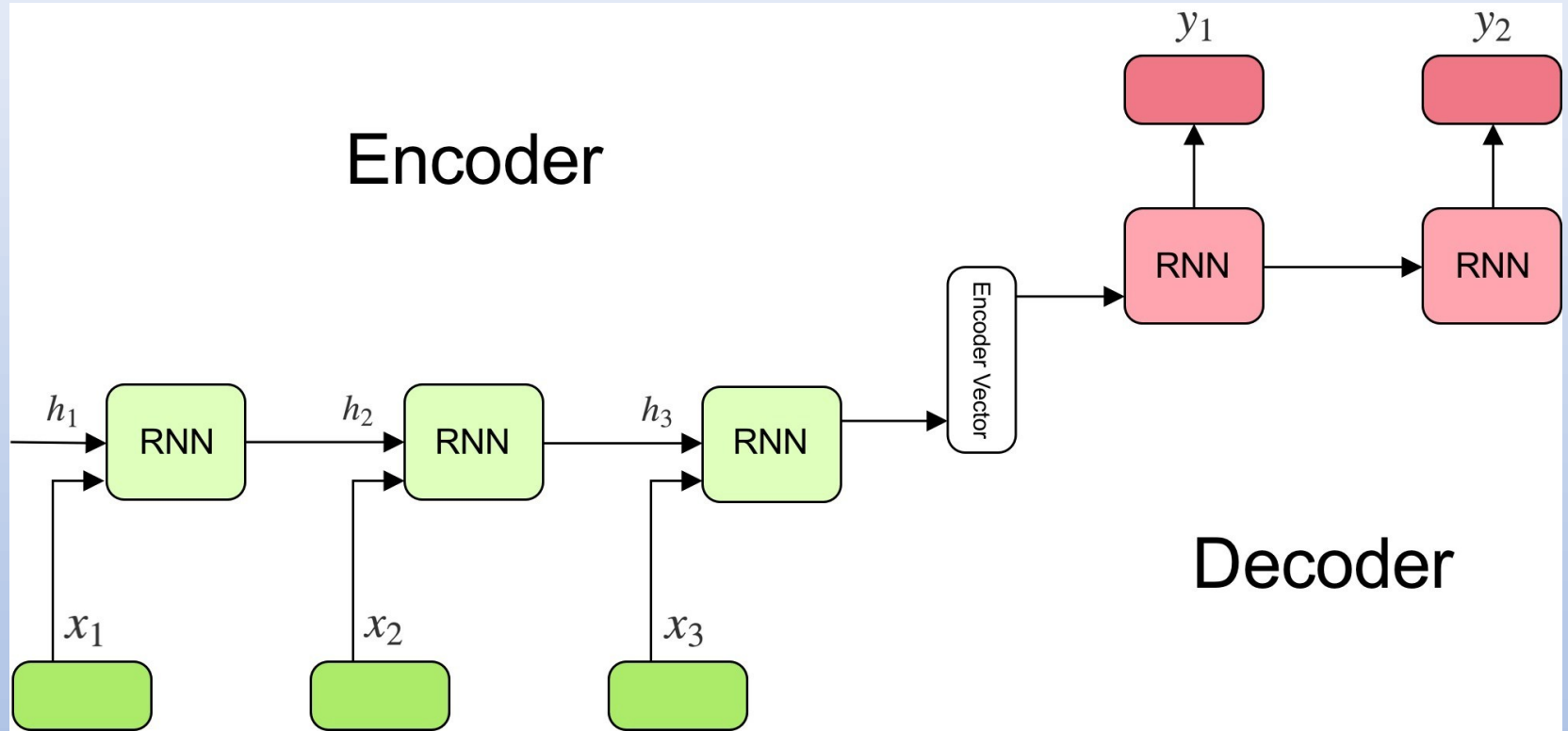
Машинный перевод

Attention is all you need

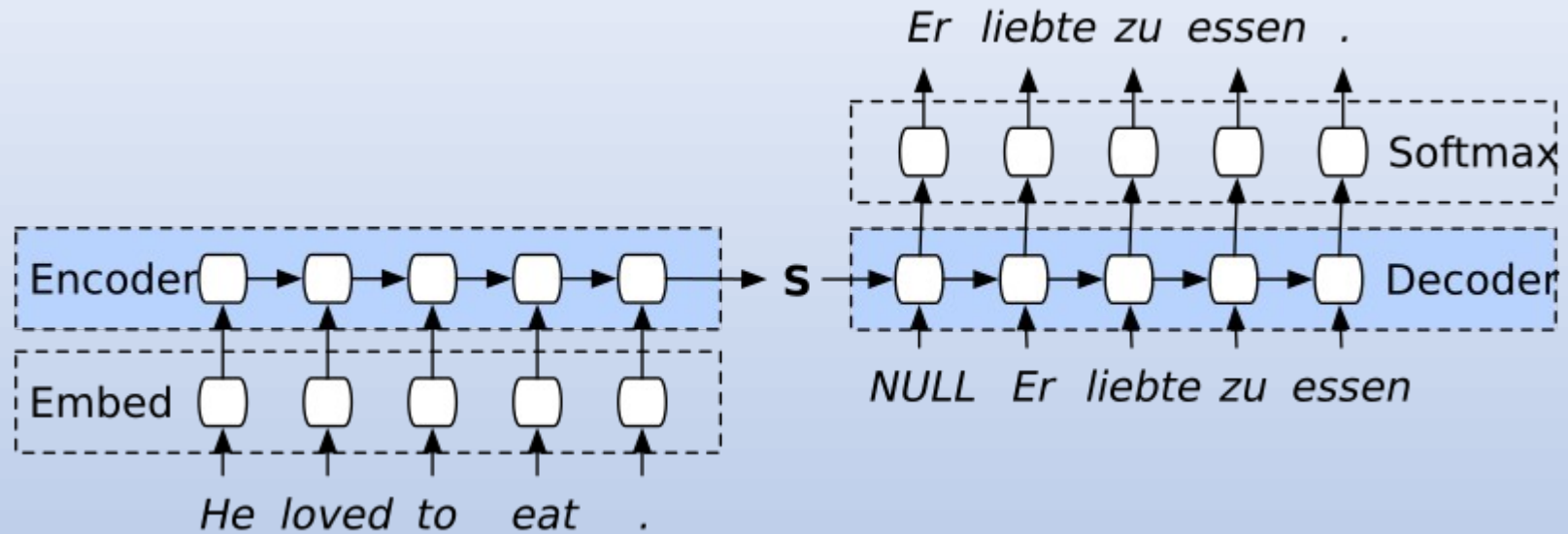
<https://jalammar.github.io/illustrated-transformer/>



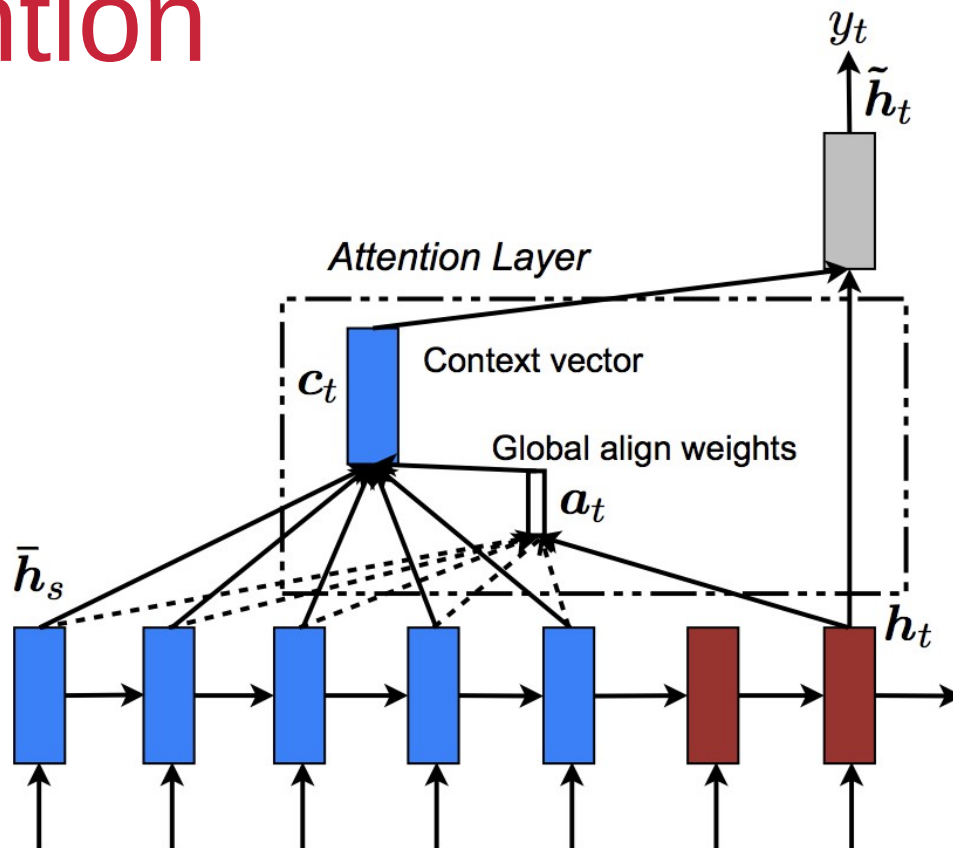
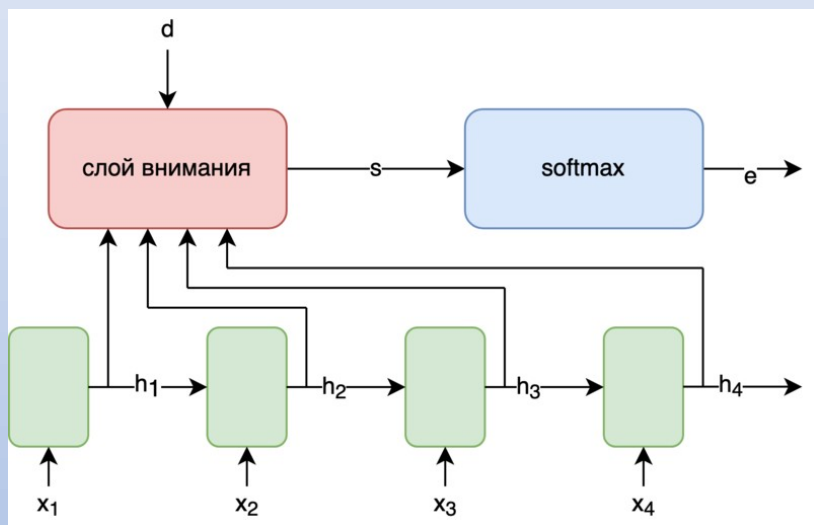
seq2seq



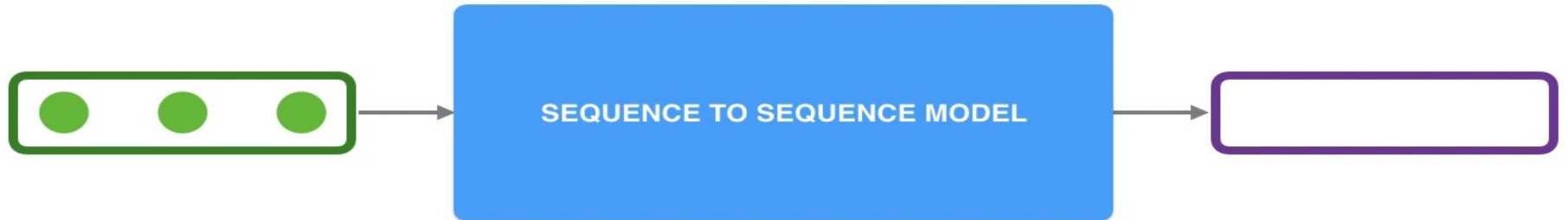
seq2seq



Seq2seq with attention

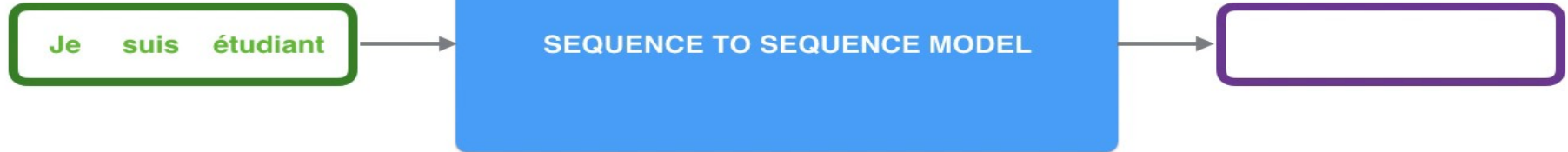


Seq2seq model

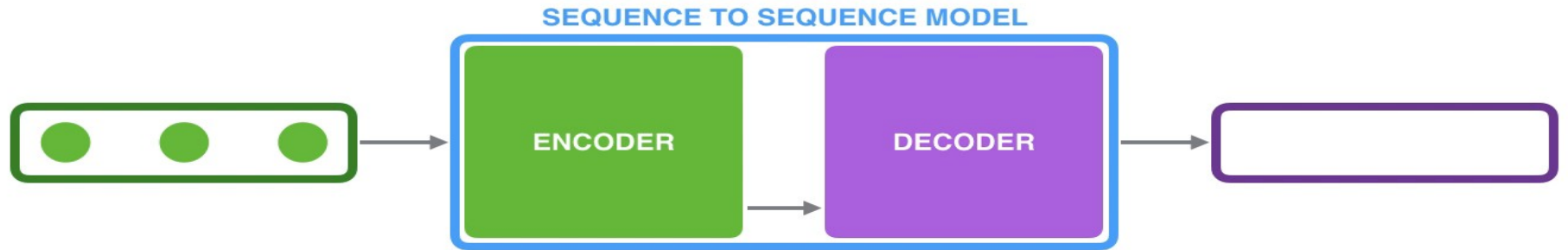


Seq2seq: machine translation

Neural Machine Translation SEQUENCE TO SEQUENCE MODEL

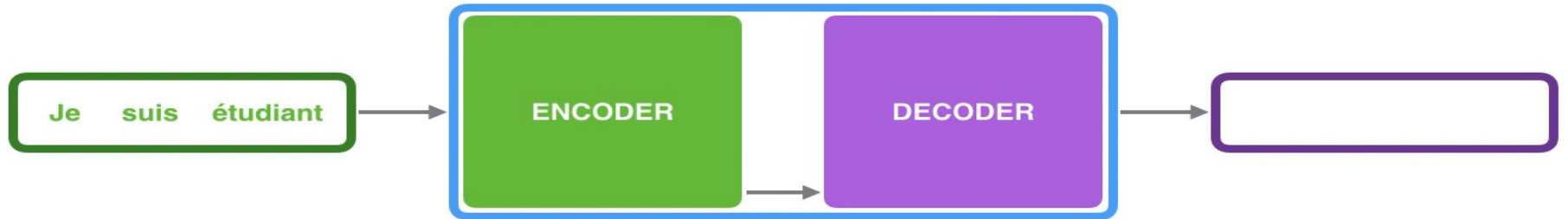


Seq2seq: encoder - decoder



Seq2seq: encoder - decoder

Neural Machine Translation SEQUENCE TO SEQUENCE MODEL



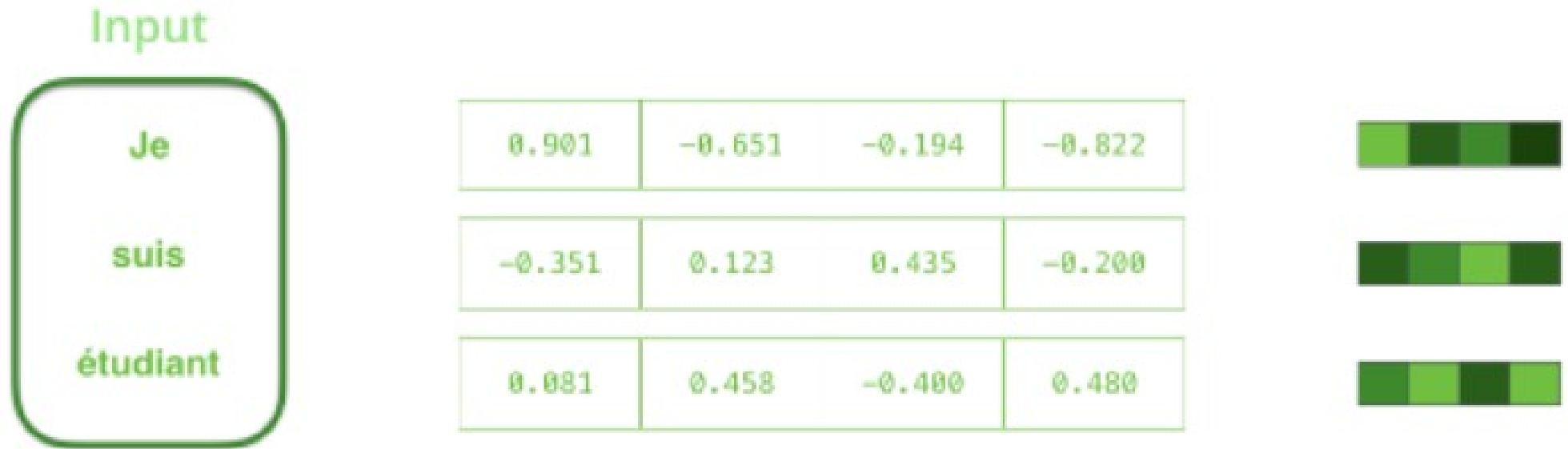
Seq2seq: contex

CONTEXT

0.11
0.03
0.81
-0.62

0.11
0.03
0.81
-0.62

Seq2seq: input for encoder



RNN

Recurrent Neural Network

Time step #1:

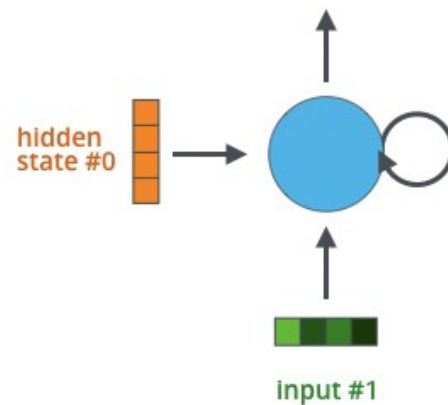
An RNN takes two input vectors:



hidden
state #0



input vector #1



Seq2seq: hidden state

Time step:

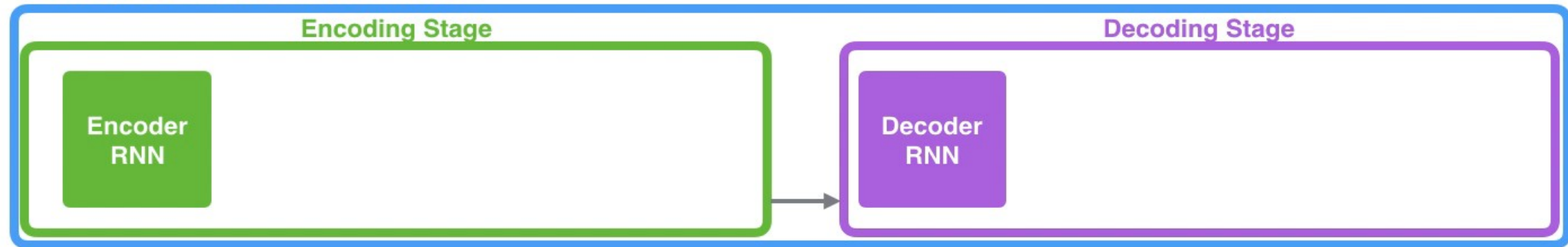
Neural Machine Translation
SEQUENCE TO SEQUENCE MODEL



Seq2seq

Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL



Je

suis

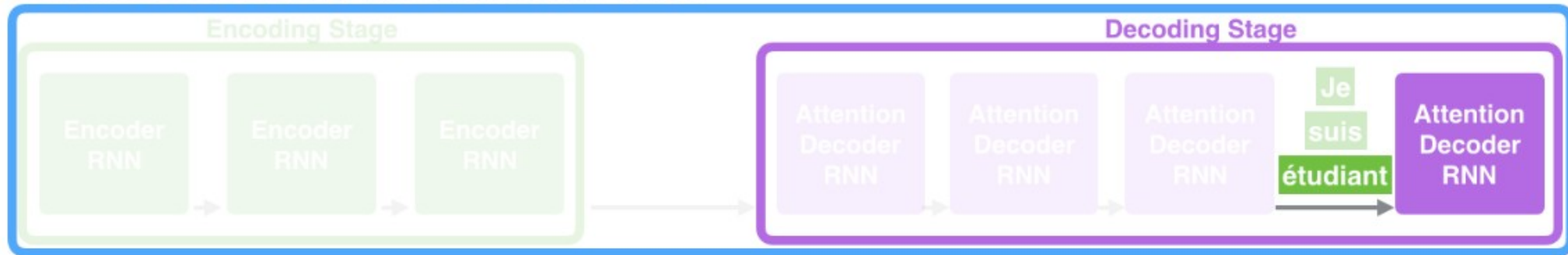
étudiant

Seq2seq: attention

Time step: 7

Neural Machine Translation

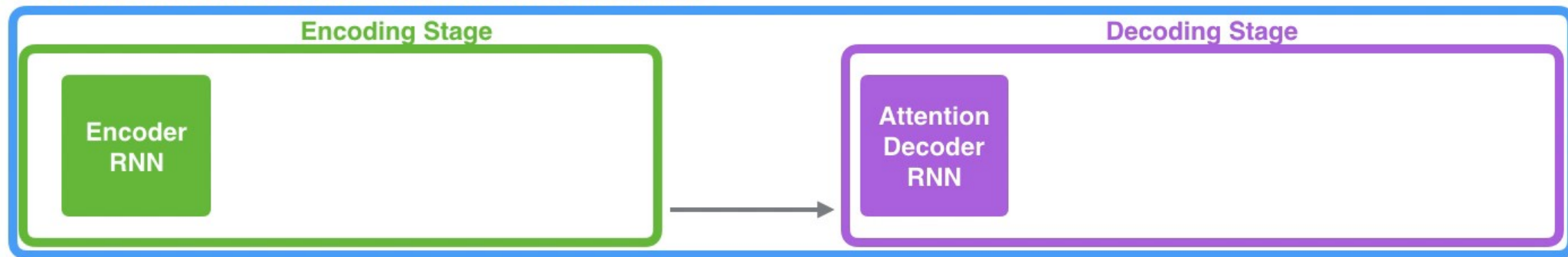
SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Seq2seq: attention

Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL WITH ATTENTION



Je

suis

étudiant

Attention at time step 4



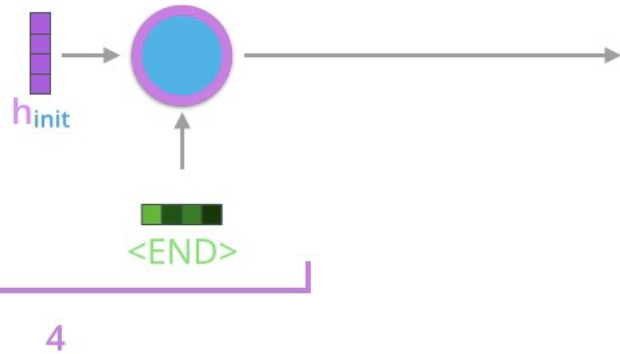
Neural Machine Translation

SEQUENCE TO SEQUENCE MODEL WITH ATTENTION

Encoding Stage



Attention Decoding Stage



Seq2seq: attention

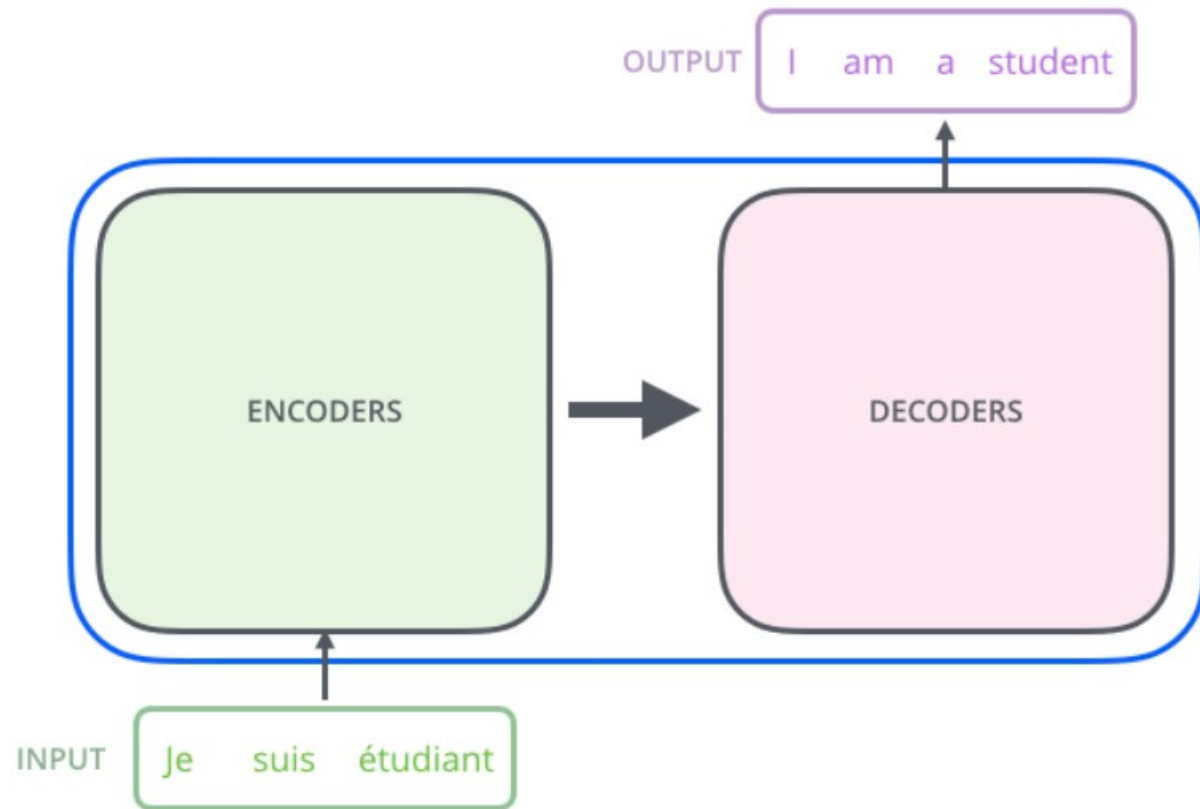




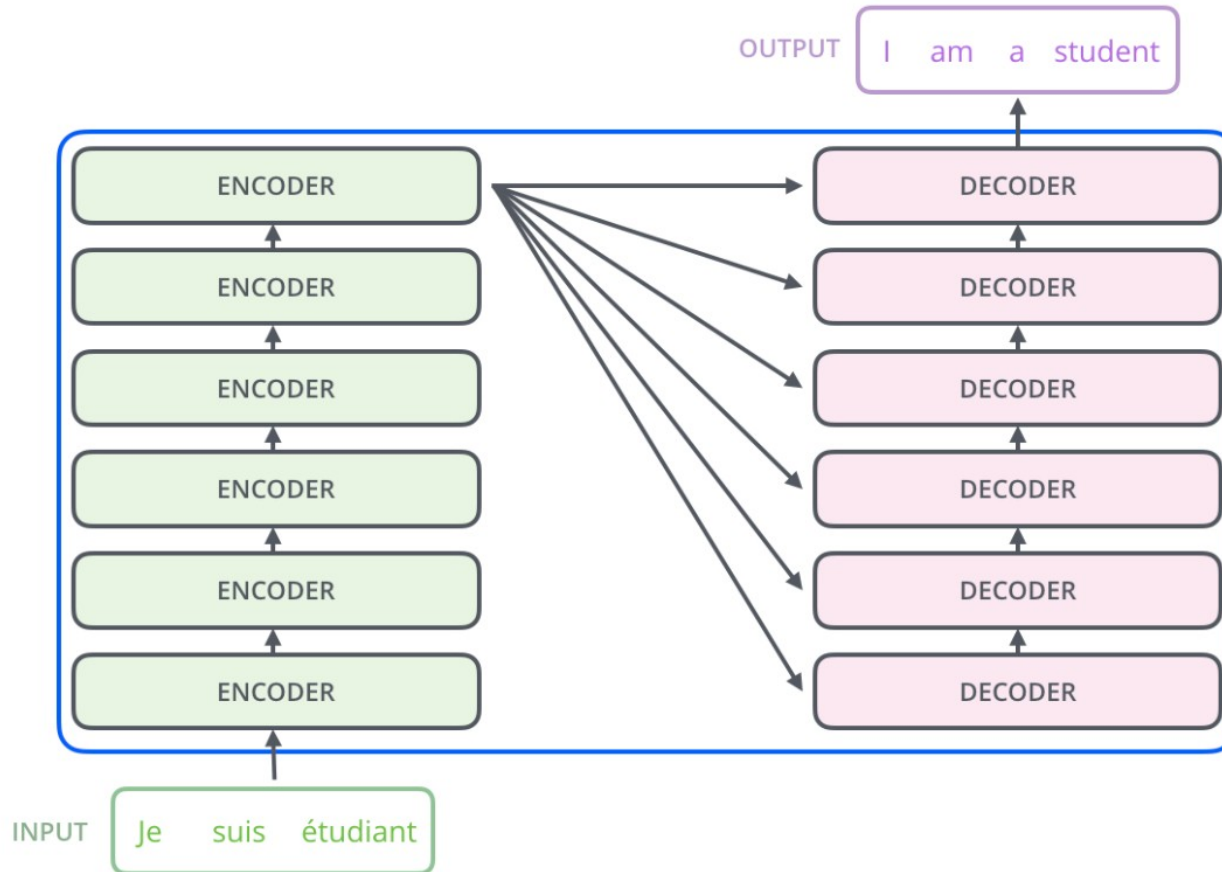
Трансформер



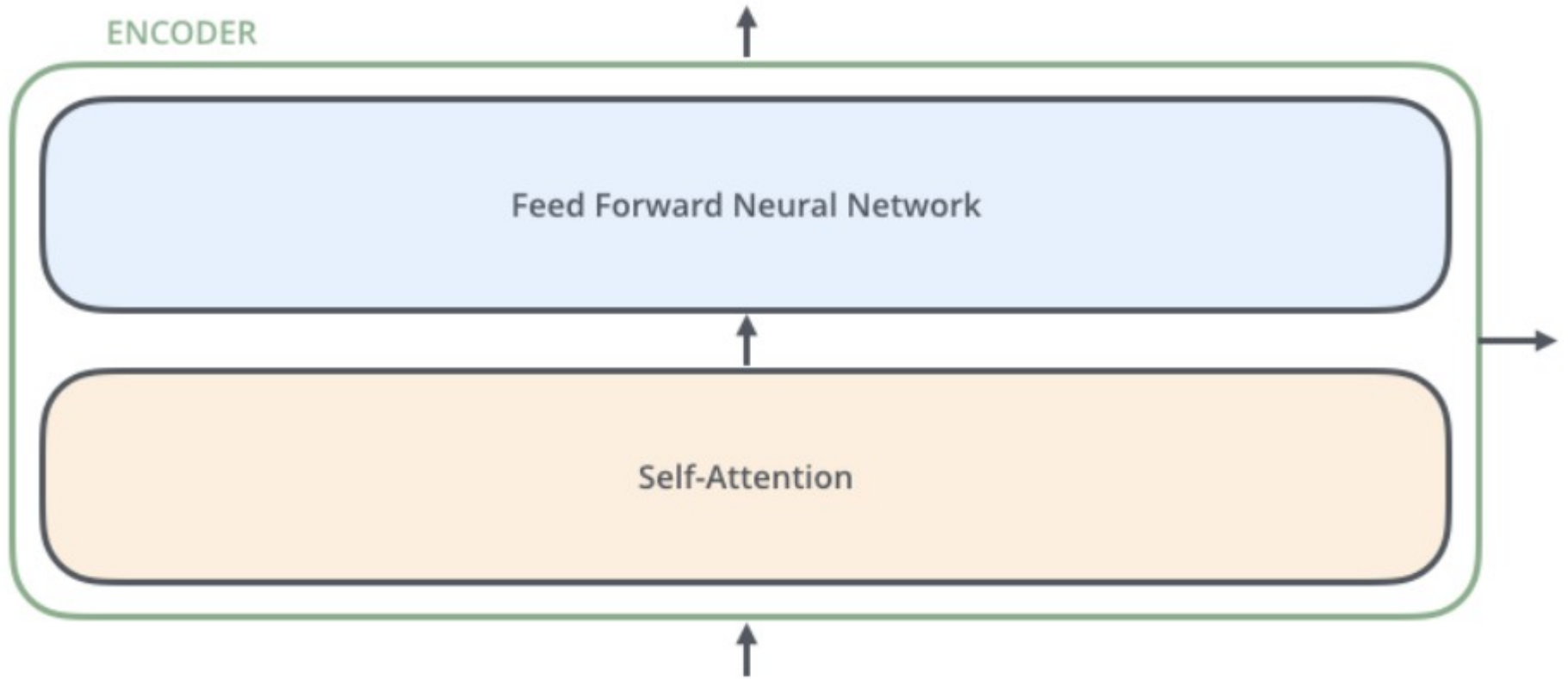
Encoder - decoder



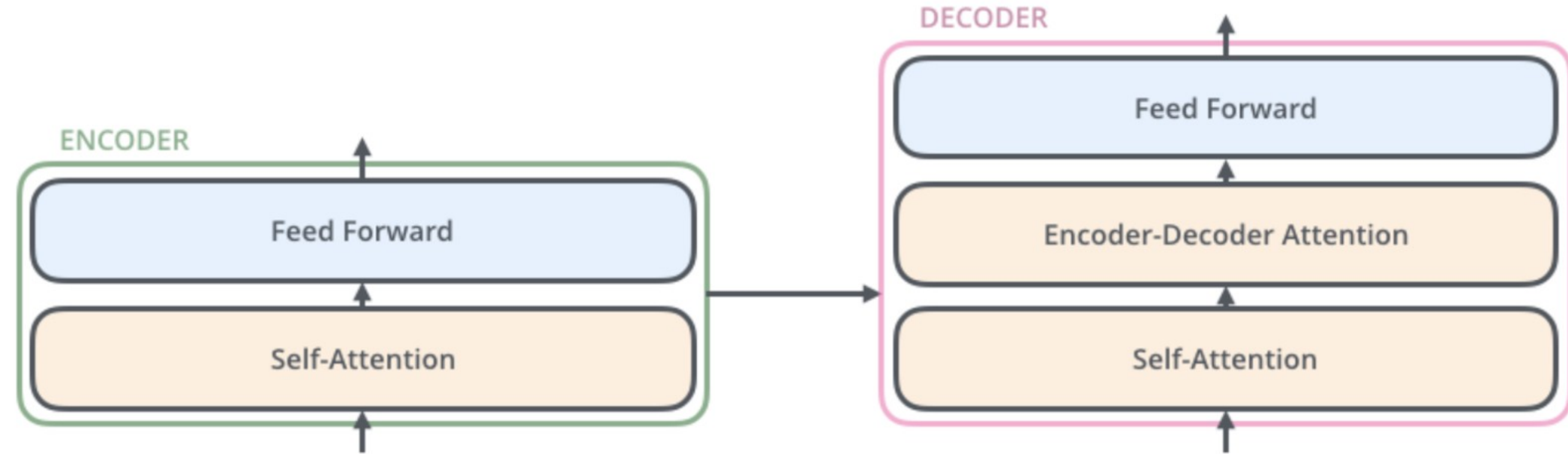
Encoder - decoder

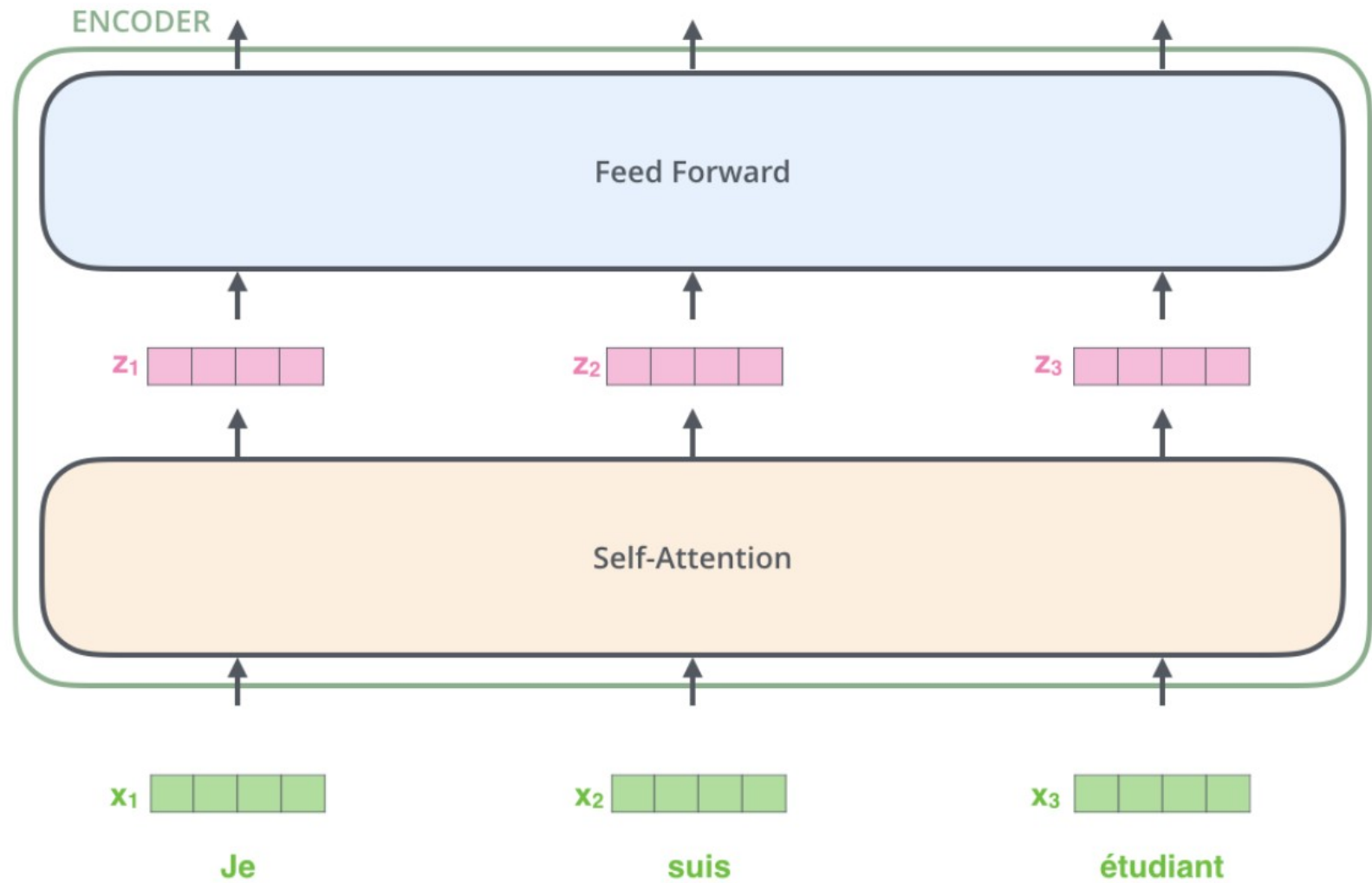


Encoder



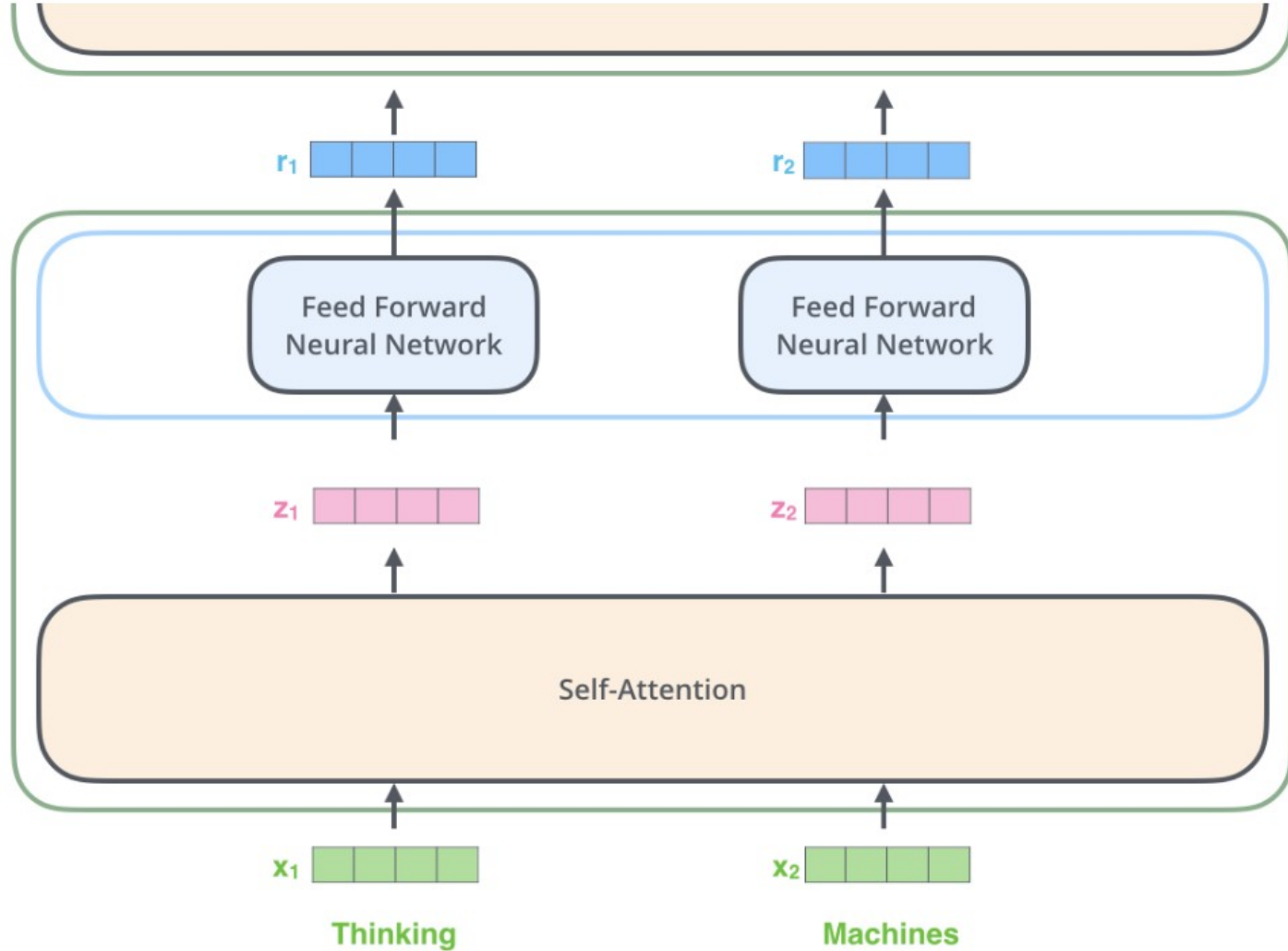
Encoder - decoder





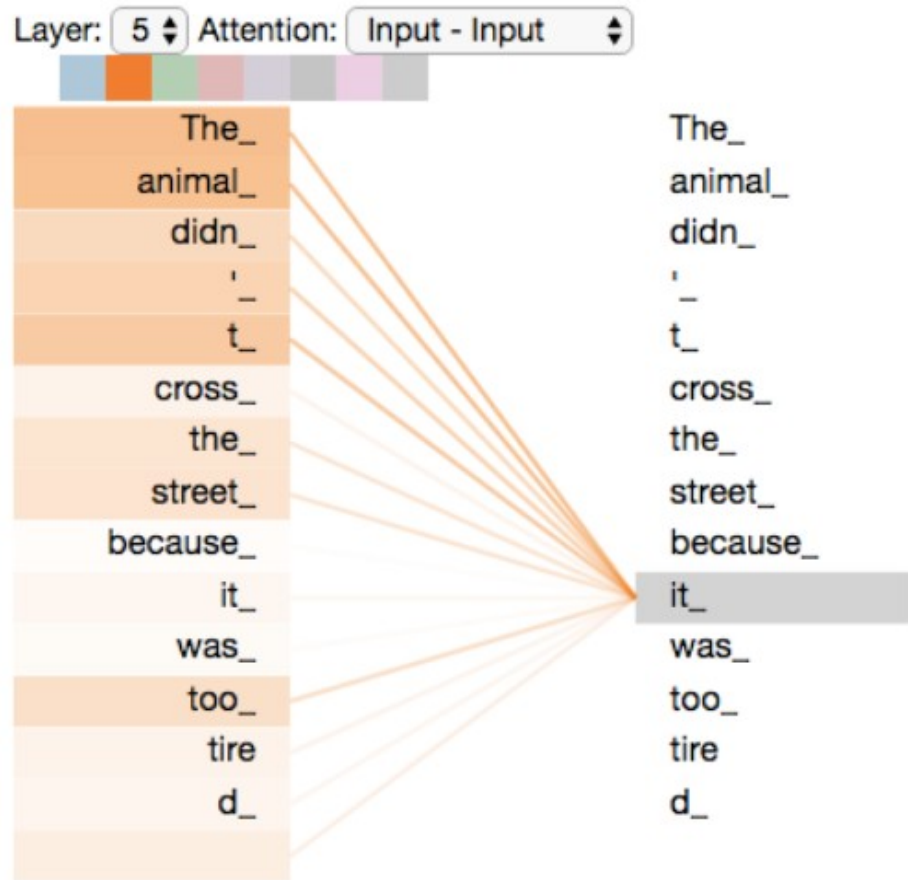
ENCODER #2

ENCODER #1



” The ani mal di dn' t cross the street
because it was too tired”

□ К чему относиться it в этом предложении (animal или street)?

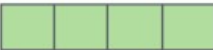


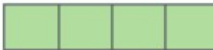
Input

Thinking

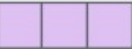
Machines

Embedding

x_1 

x_2 

Queries

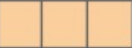
q_1 

q_2 



W^Q

Keys

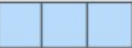
k_1 

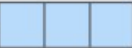
k_2 



W^K

Values

v_1 

v_2 



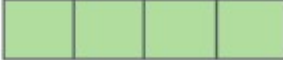
W^V

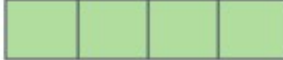
Input

Thinking

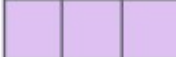
Machines

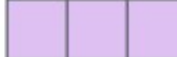
Embedding

x_1 

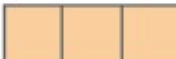
x_2 

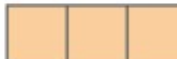
Queries

q_1 

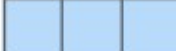
q_2 

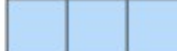
Keys

k_1 

k_2 

Values

v_1 

v_2 

Score

$$q_1 \cdot k_1 = 112$$

$$q_1 \cdot k_2 = 96$$

Input

Embedding

Queries

Keys

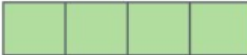
Values

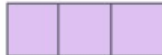
Score

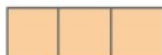
Divide by 8 ($\sqrt{d_k}$)

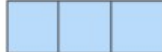
Softmax

Thinking

x_1 

q_1 

k_1 

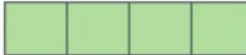
v_1 

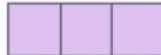
$$q_1 \cdot k_1 = 112$$

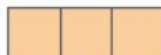
14

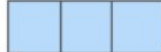
0.88

Machines

x_2 

q_2 

k_2 

v_2 

$$q_1 \cdot k_2 = 96$$

12

0.12

Input

Embedding

Queries

Keys

Values

Score

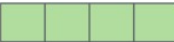
Divide by $8 (\sqrt{d_k})$

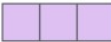
Softmax

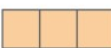
Softmax
X
Value

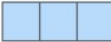
Sum

Thinking

x_1 

q_1 

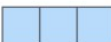
k_1 

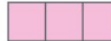
v_1 

$q_1 \cdot k_1 = 112$

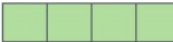
14

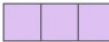
0.88

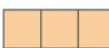
v_1 

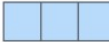
z_1 

Machines

x_2 

q_2 

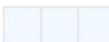
k_2 

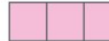
v_2 

$q_2 \cdot k_2 = 96$

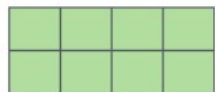
12

0.12

v_2 

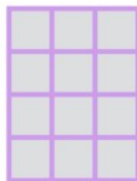
z_2 

X



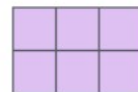
$\times$

W^Q

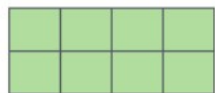


$=$

Q

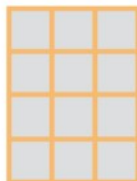


X



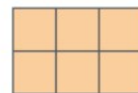
$\times$

W^K

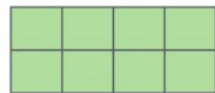


$=$

K



X



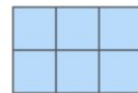
$\times$

W^V



$=$

V



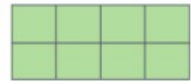
$$\text{softmax} \left(\frac{\overset{\text{Q}}{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} \times \overset{\text{K}^T}{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array}} \right) \overset{\text{V}}{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}}$$

$\sqrt{d_k}$

$$= \overset{\text{Z}}{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}}$$

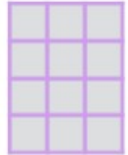
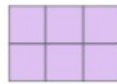
X

Thinking
Machines



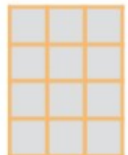
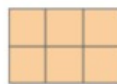
ATTENTION HEAD #0

Q_0



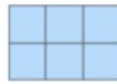
W_0^Q

K_0



W_0^K

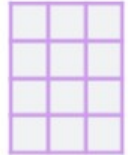
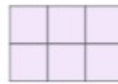
V_0



W_0^V

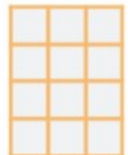
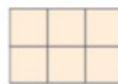
ATTENTION HEAD #1

Q_1



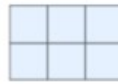
W_1^Q

K_1



W_1^K

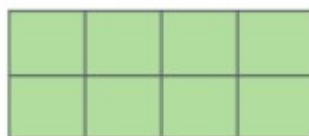
V_1



W_1^V

X

Thinking
Machines

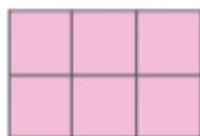


Calculating attention separately in
eight different attention heads



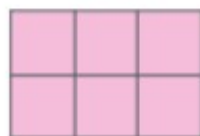
ATTENTION
HEAD #0

Z_0



ATTENTION
HEAD #1

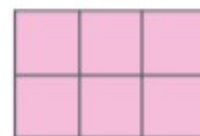
Z_1



...

ATTENTION
HEAD #7

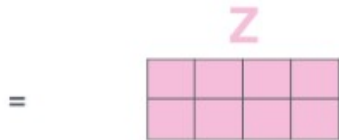
Z_7



1) Concatenate all the attention heads

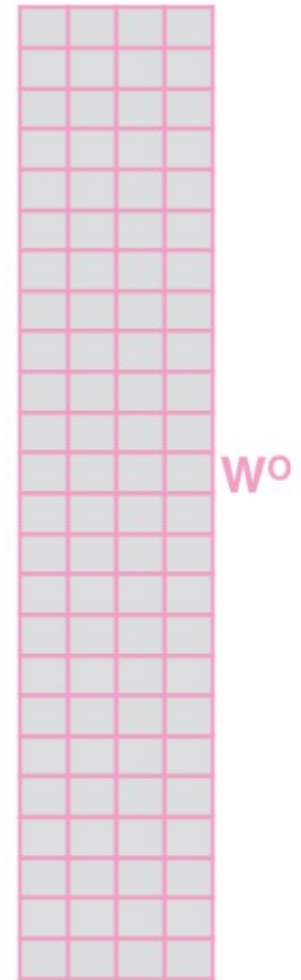


3) The result would be the Z matrix that captures information from all the attention heads. We can send this forward to the FFNN



2) Multiply with a weight matrix W^O that was trained jointly with the model

$\times$



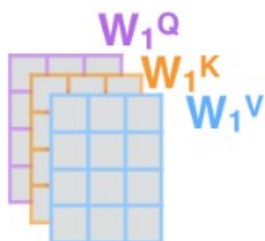
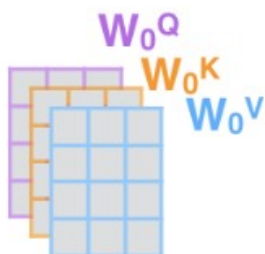
1) This is our input sentence*

Thinking
Machines

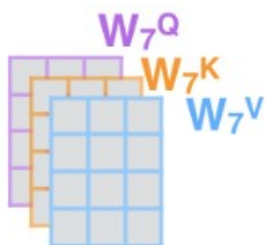
2) We embed each word*



3) Split into 8 heads.
We multiply X or R with weight matrices



...



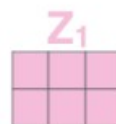
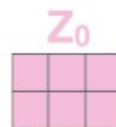
4) Calculate attention using the resulting $Q/K/V$ matrices



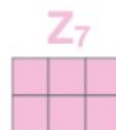
...



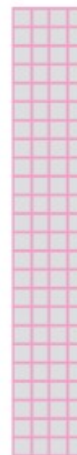
5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer



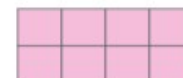
...



W^O



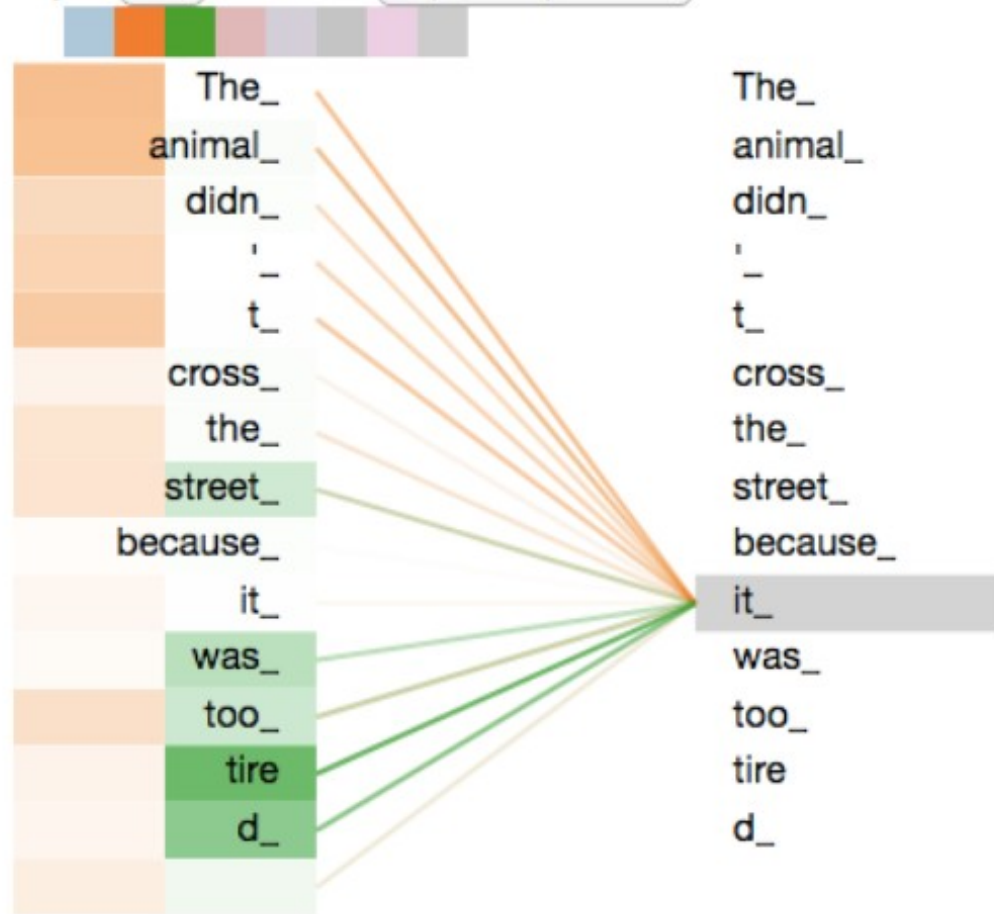
Z



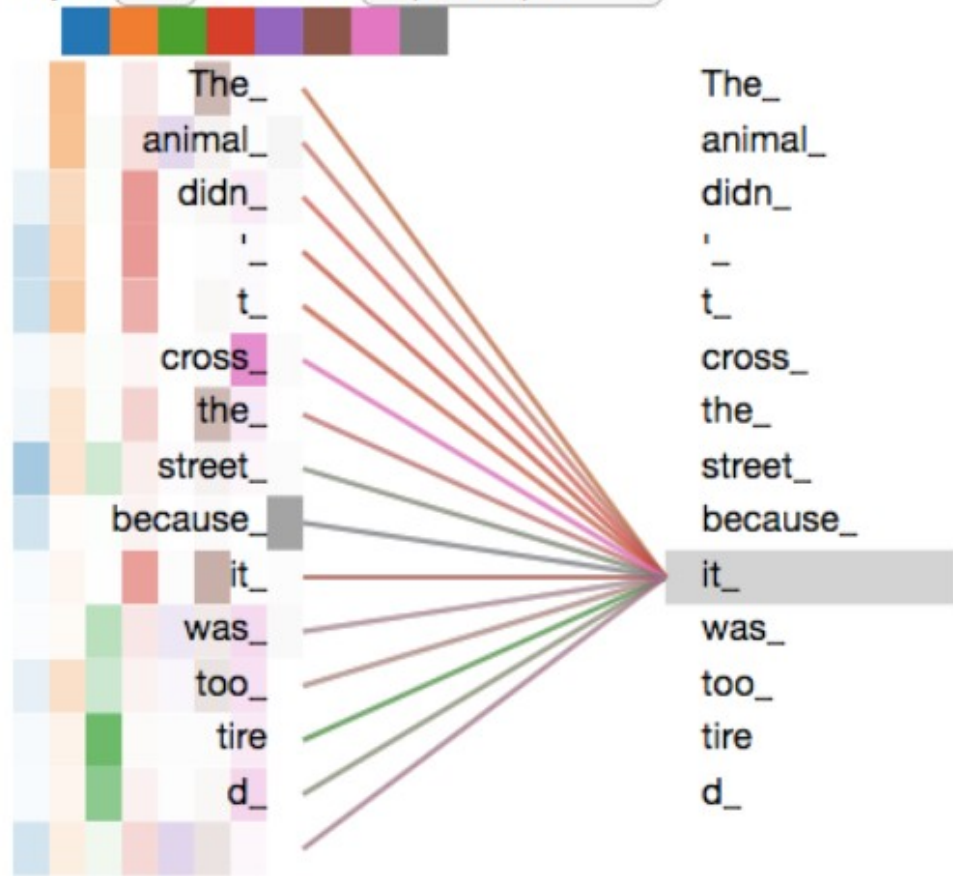
* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one

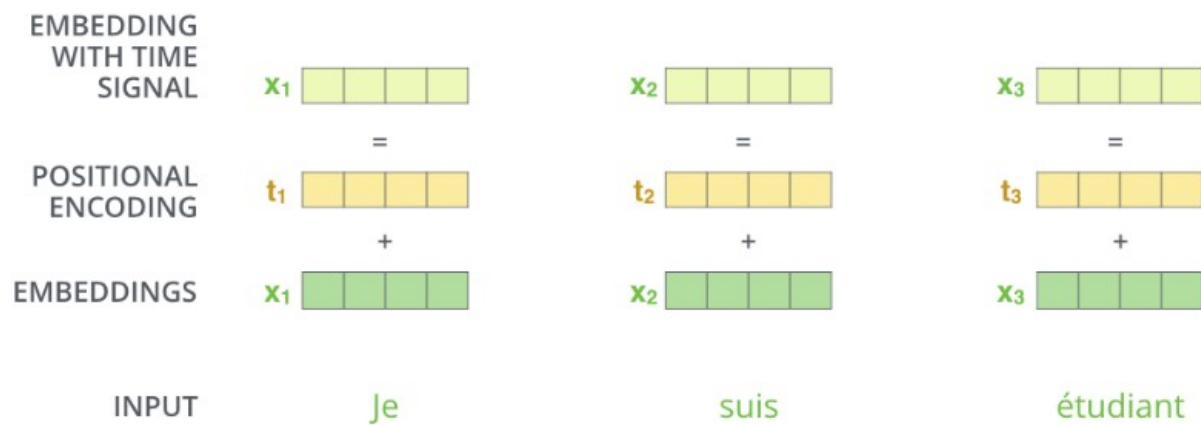
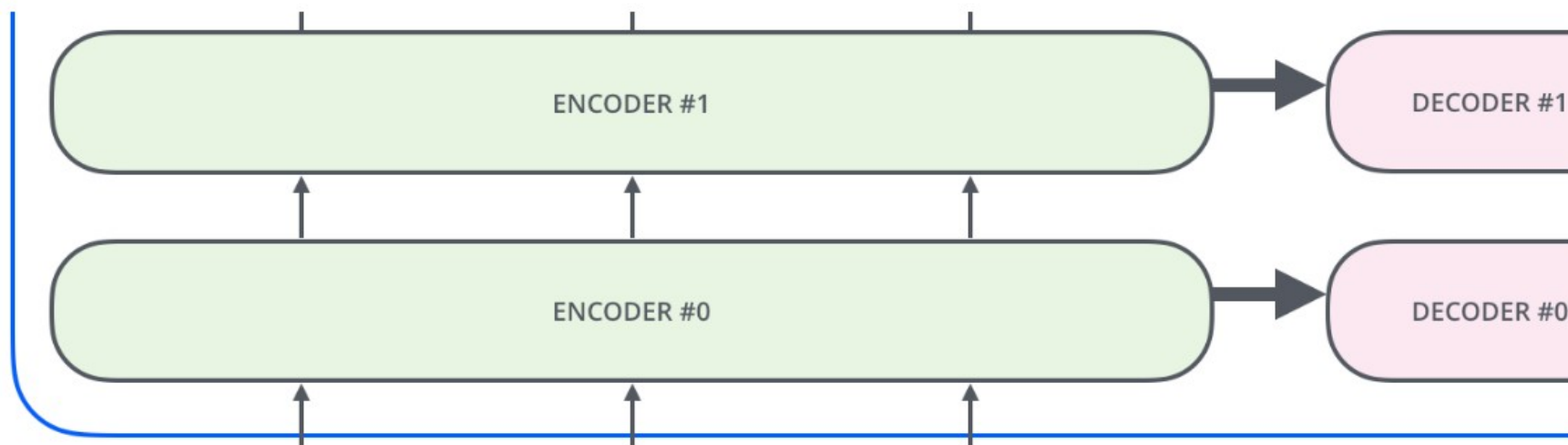


Layer: 5 Attention: Input - Input

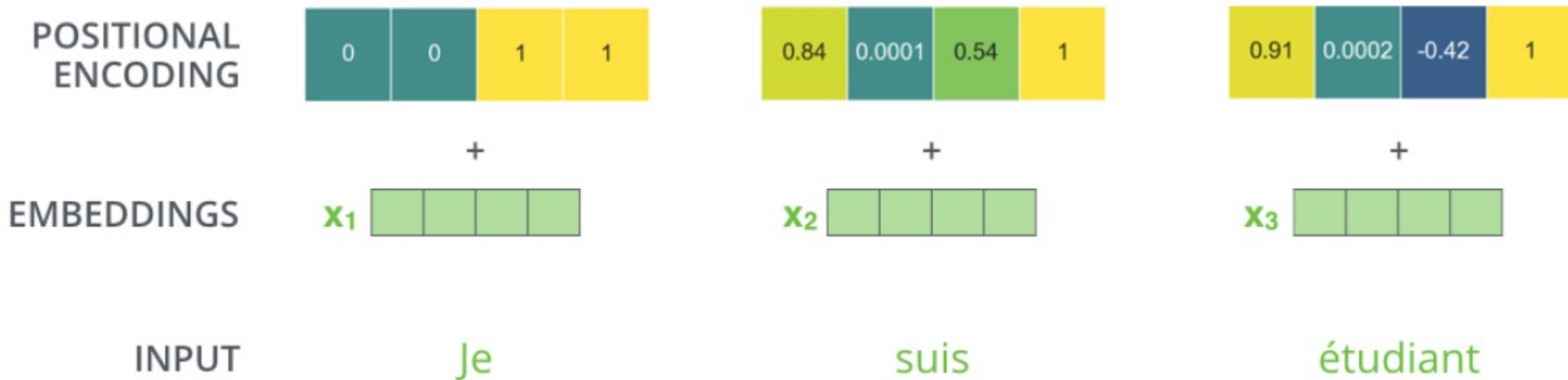


Layer: 5 Attention: Input - Input





Positional encoding



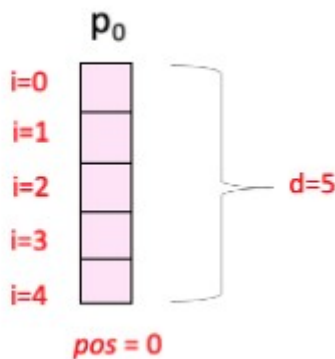
https://github.com/jalamar/jalamar.github.io/blob/master/not_ebooks/transformer_transformer_positional_encoding_graph.ipynb

Positional encoding

$$PE_{(pos, 2i)} = \sin(pos / 10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i/d_{model}})$$

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right)$$

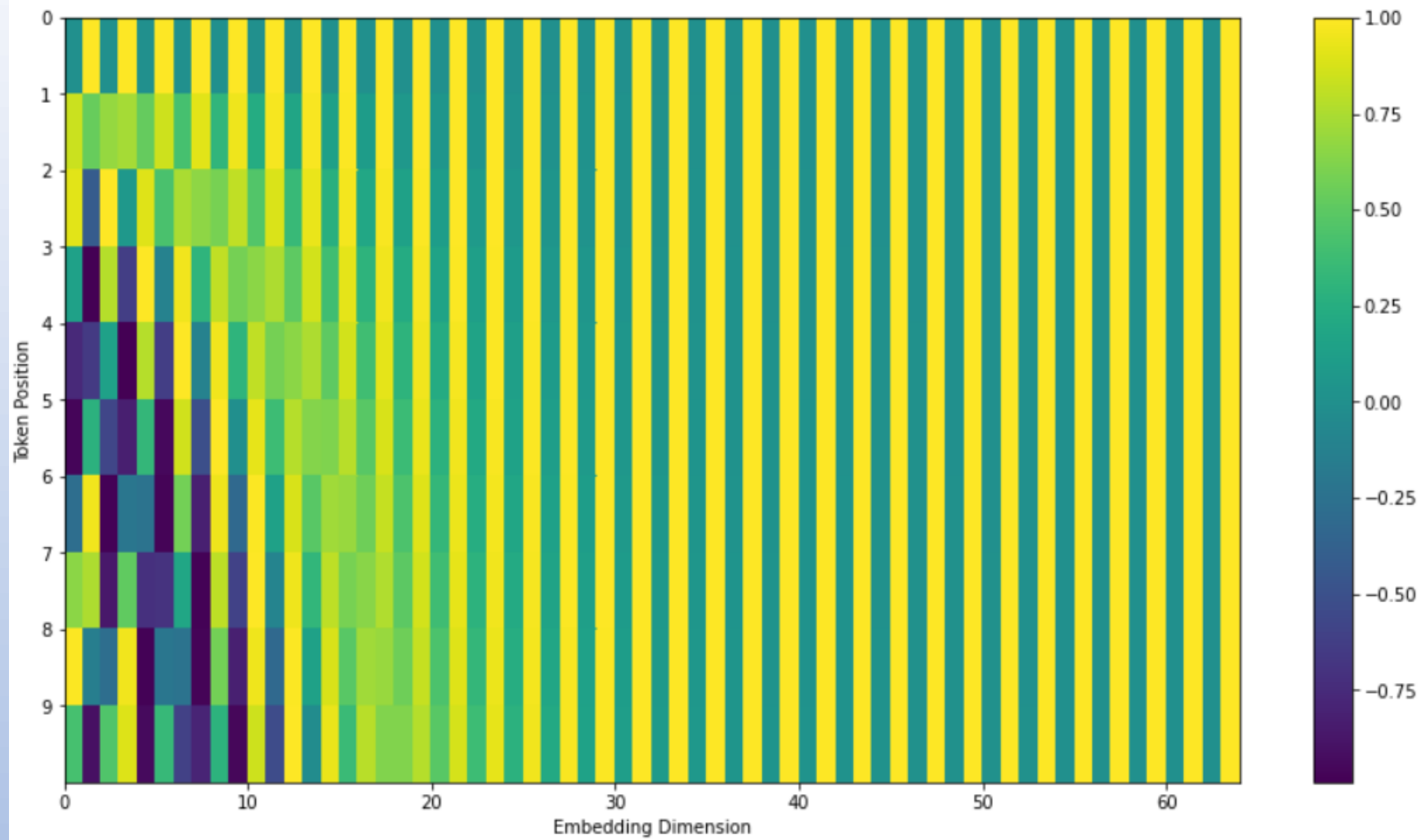


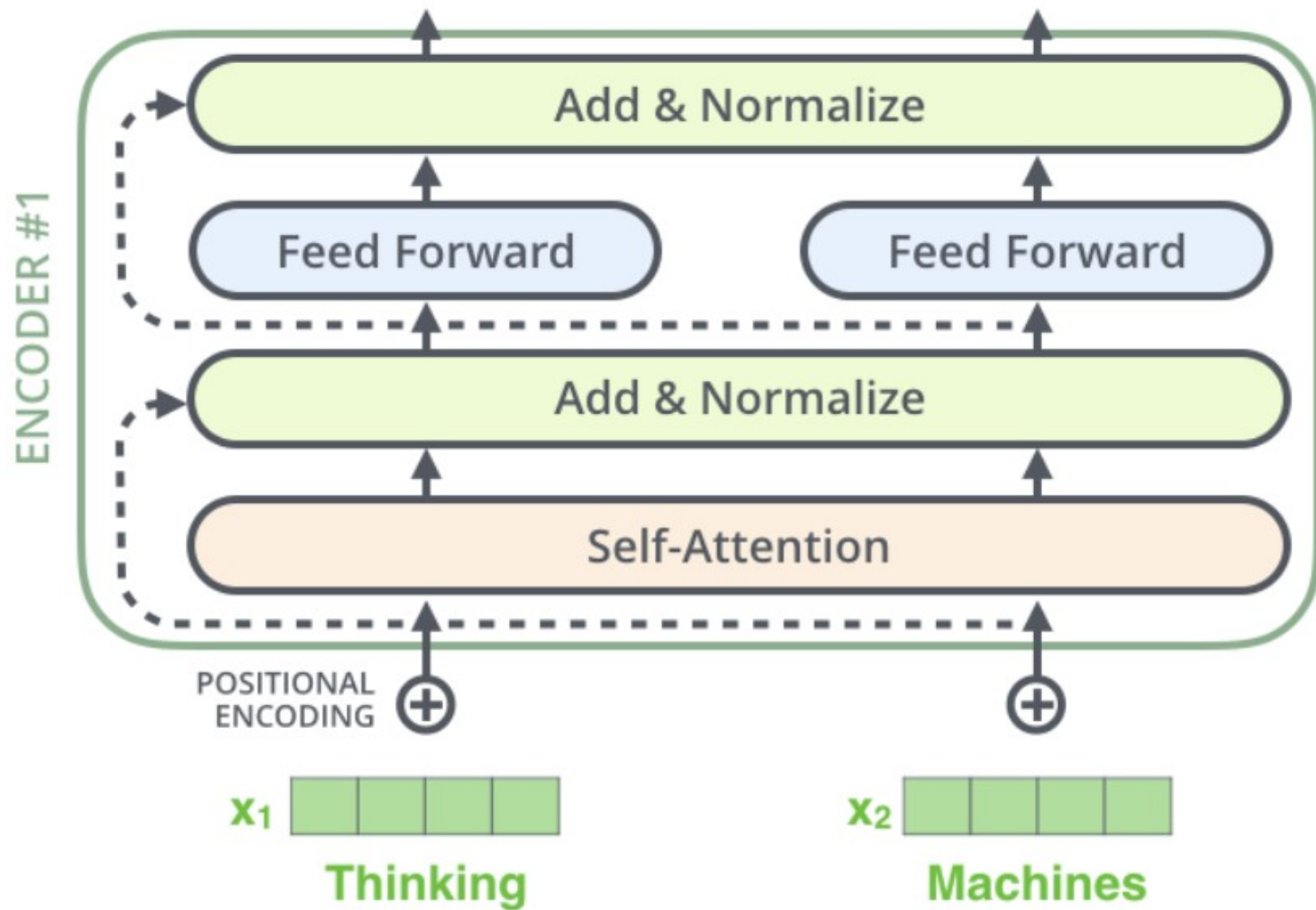
$$\vec{p}_t = \begin{bmatrix} \sin(\omega_1.t) \\ \cos(\omega_1.t) \\ \\ \sin(\omega_2.t) \\ \cos(\omega_2.t) \\ \\ \vdots \\ \\ \sin(\omega_{d/2}.t) \\ \cos(\omega_{d/2}.t) \end{bmatrix}_{d \times 1}$$

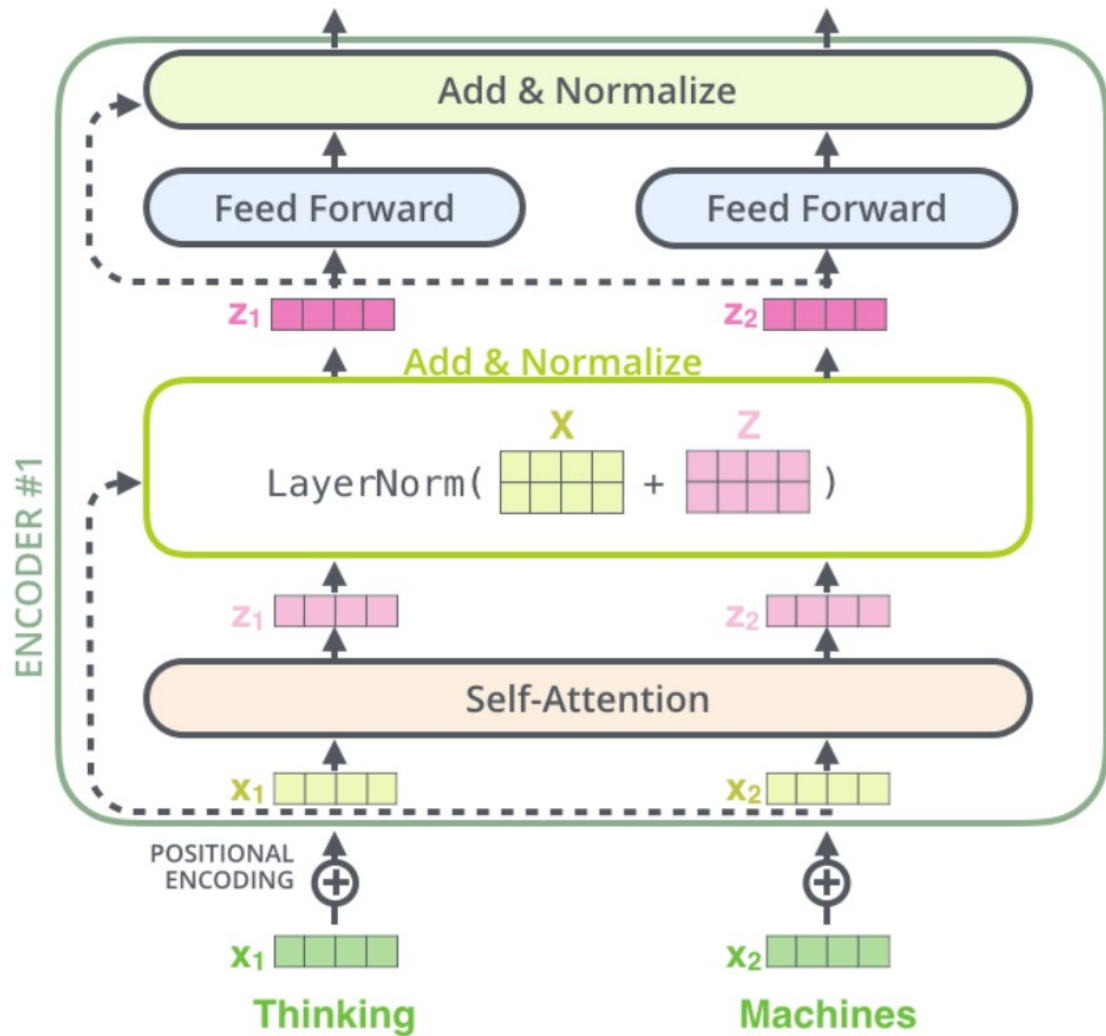
$$u_1 = \cos(\omega_k.\phi) \quad v_1 = \sin(\omega_k.\phi) \quad (1)$$

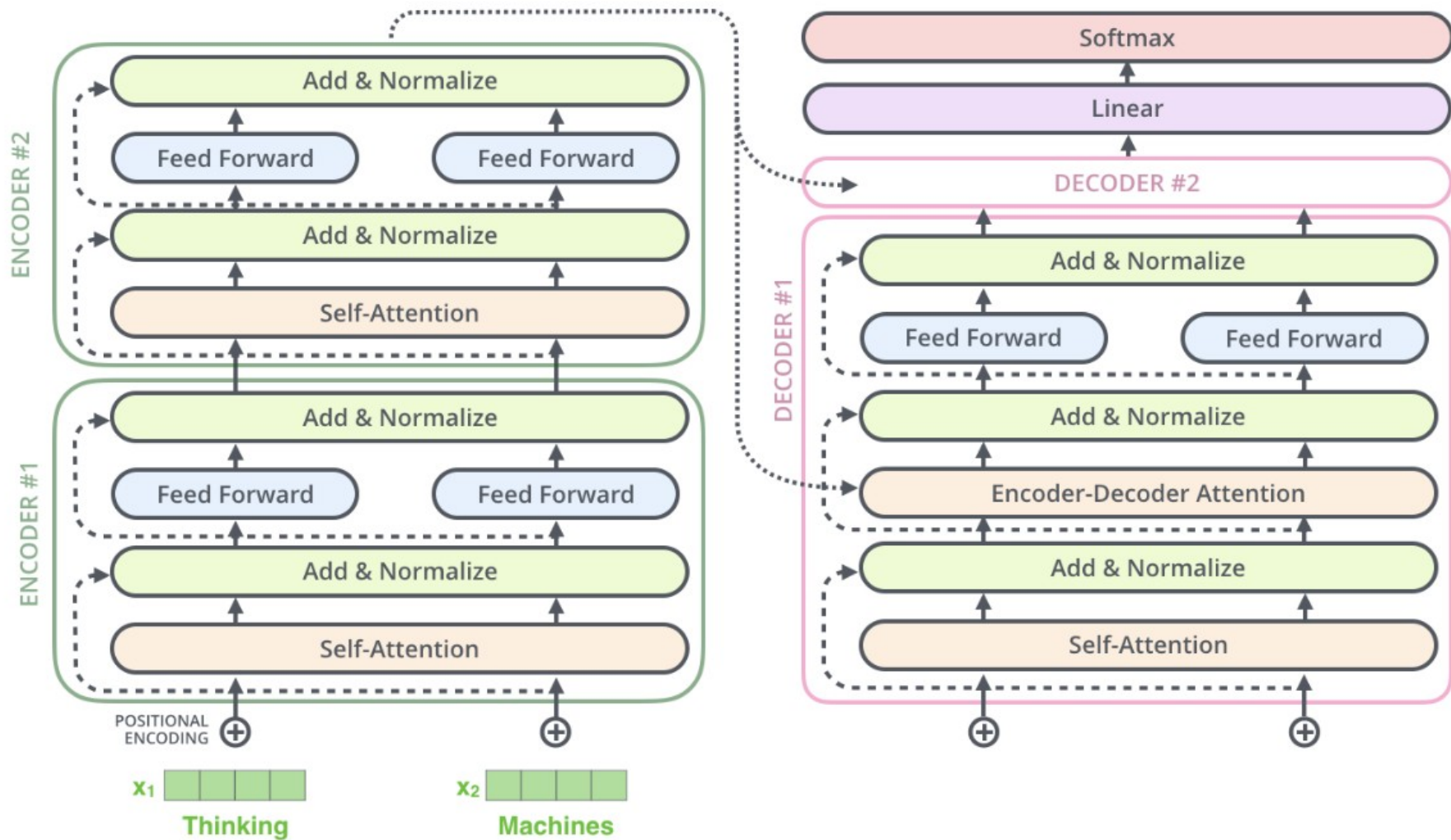
$$u_2 = -\sin(\omega_k.\phi) \quad v_2 = \cos(\omega_k.\phi) \quad (2)$$

$$\begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix} \cdot \begin{bmatrix} \sin(\omega_k.t) \\ \cos(\omega_k.t) \end{bmatrix} = \begin{bmatrix} \sin(\omega_k.(t + \phi)) \\ \cos(\omega_k.(t + \phi)) \end{bmatrix}$$



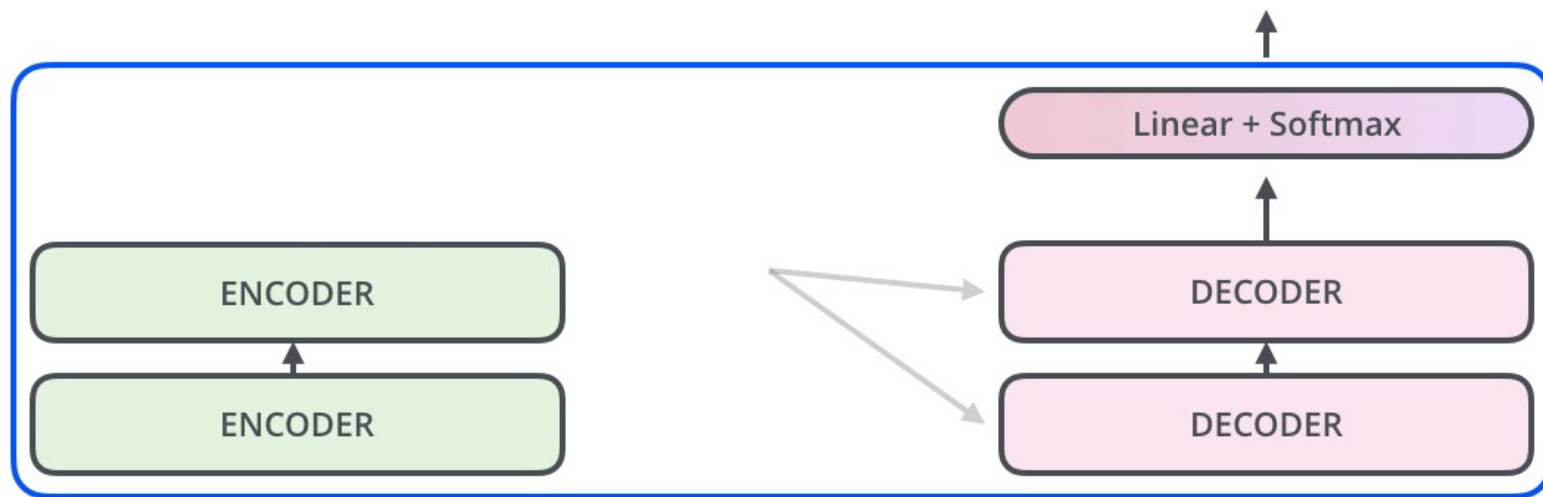




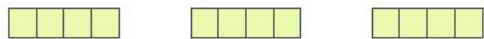


Decoding time step: 1 2 3 4 5 6

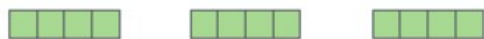
OUTPUT



EMBEDDING
WITH TIME
SIGNAL



EMBEDDINGS



INPUT

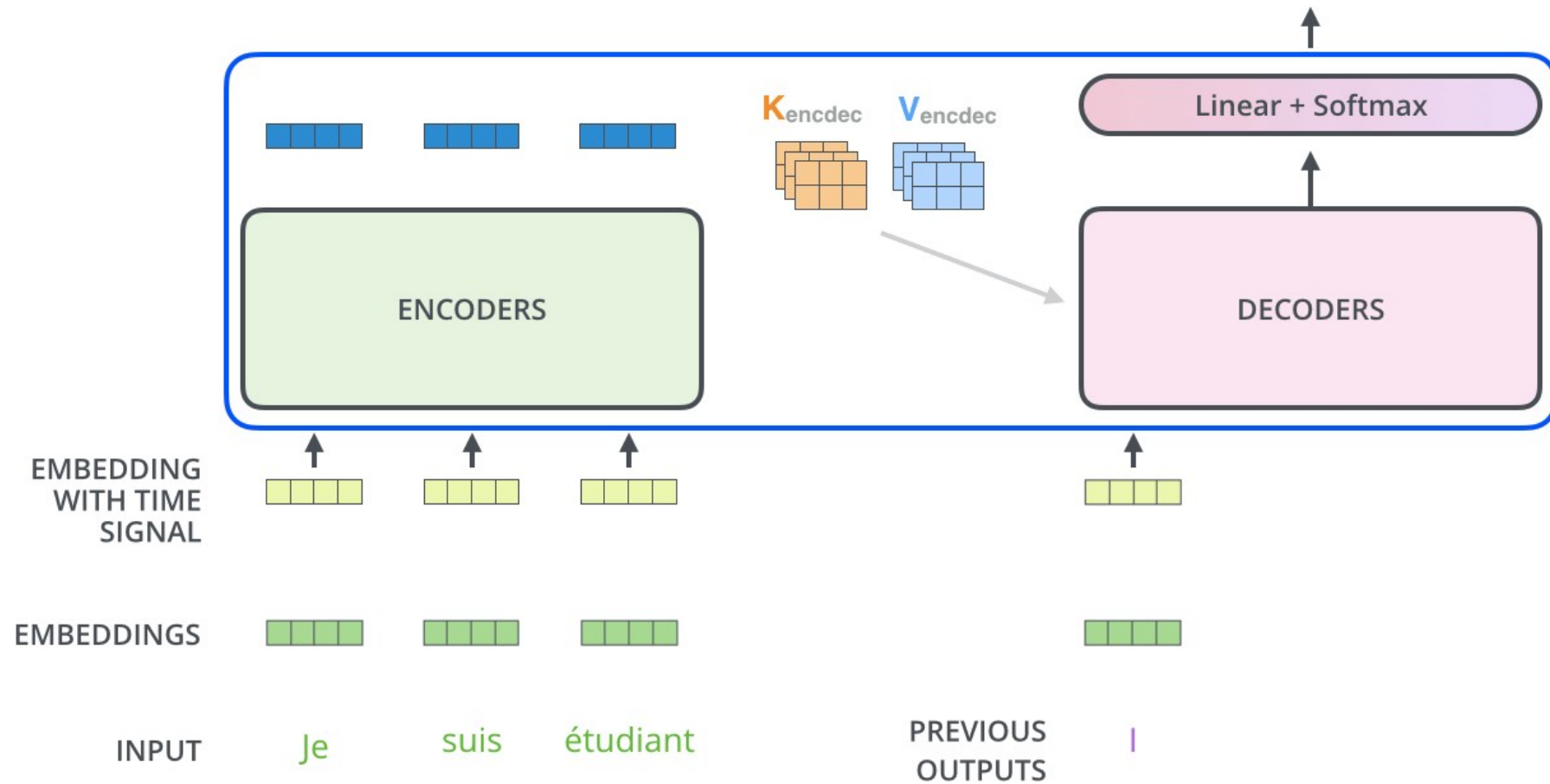
Je

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Decoding time step: 1 2 3 4 5 6

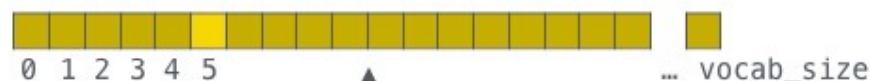
OUTPUT



Which word in our vocabulary
is associated with this index?

Get the index of the cell
with the highest value
(**argmax**)

log_probs

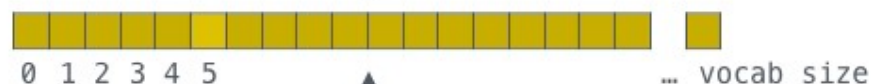


am

5

Softmax

logits



Linear

Decoder stack output

